

__TITLE__

__SUBTITLE__

Audience: __AUDIENCE__ | **PLC:** __PLC_MODEL_AND_PART__

Drive: __DRIVE_MODEL_AND_ADDRESS__

Project: __PROJECT_FILE__ | **IDE:** __IDE_VERSION__

Primary reference: __PRIMARY_MANUAL__

This doc supersedes: __PRIOR_DOC_OR_NONE__

You will understand after reading: __LEARNING_OUTCOME_ONE_SENTENCE__

① One-page cheat sheet (the values that absolutely must be right)

What	Value	Source
__VALUE_NAME__	__VALUE__	__SOURCE_CITATION__

⚠ __N__ bugs in the current __PRIOR_ARTIFACT__ that this doc fixes

1. __BUG_HEADLINE__: __DESCRIPTION_WITH_WRONG_AND_CORRECT_VALUES__

1 The Big Picture

__WHAT_WE_ARE_TRYING_TO_ACCOMPLISH_IN_PLAIN_LANGUAGE__

2 The Real-World Problem

__THE_PHYSICAL_MACHINE_SCENARIO_THE_READER_SHOULD_PICTURE__

3 The Simple Version First

__INTRO_TO_THE_DUMB_VERSION_THAT_WORKS__

__LADDER_OR_ST_OF_THE_SIMPLE_VERSION__

__LINE_BY_LINE_WALKTHROUGH__

If you see this, the simple version works

__OBSERVABLE_PROOF_THE_SIMPLE_VERSION_RUNS__

4 Why This Becomes a Problem

__WHAT_HAPPENS_WHEN_THE_SIMPLE_VERSION_IS_COPIED_THREE_TIMES__

The pain made concrete

__NAMED_SCENARIO_THE_READER_SHOULD_FEEL_BAD_ABOUT__

5 The Better Pattern

__INTRODUCE_THE_PATTERN_BY_NAME_AND_EXPLAIN_IT_IN_PLAIN_ENGLISH__

Why this matters: __SHORT_VIVID_ANALOGY_THAT_ANCHORS_THE_PATTERN__

6 CCW Step-by-Step Walkthrough

__INTRO_TO_THE_HANDS_ON_PART__

- ☐ __FIRST_CLICK_OR_ACTION_WITH_EXPECTED_RESULT__
- ☐ __SECOND_CLICK_OR_ACTION__

The "no match" empty dropdown is fine

__ANTICIPATE_THE_CCW_PANIC_FOR_THIS_STEP_FROM_BEGINNER_VOICE_MD__

7 Variable Map

Every meaningful variable used in this lesson. The "Why it belongs here" column is the teaching content — read those cells if you want to understand the design, not just memorize it.

Name	Type	Direction	Scope	Purpose	Why it belongs here
__VAR_NAME__	__TYPE__	__INPUT_OR_OUTPUT_OR_DASH__	__SCOPE_FROM_VARIABLE_MAP_MD__	__PURPOSE_IN_PLAIN_ENGLISH__	__WHY_IT_IS_IN_THIS_SCOPE_NOT_A

8 Code Walkthrough

__CHUNK_TITLE__

__LADDER_OF_THIS_CHUNK__

__WHAT_THE_PLC_SEES_AND_WHAT_CHANGES_STATE_ON_THE_NEXT_SCAN__

Why this matters: __WHAT_THIS_CHUNK_ACCOMPLISHES_MECHANICALLY__

9 Scan Cycle Explanation

Every PLC scan is the same five-phase loop:

- Read physical inputs.
- Execute the program logic.
- Update internal variables.
- Write physical outputs.
- Repeat.

__APPLY_THESE_FIVE_PHASES_TO_THE_CODE_FROM_SECTION_8__

10 Testing Procedure

- ☐ __WHAT_TO_FORCE_OR_TOGGLE_FIRST__

✓**You should see:** __OBSERVABLE_SIGNAL_THAT_CONFIRMS_THE_TEST_PASSED__

Pass / Fail summary

Test	Pass	Fail	Notes
__TEST_DESCRIPTION__	☐	☐	

11 Troubleshooting Guide

Symptom	Likely cause	What to check	Fix
__SYMPTOM__	__CAUSE__	__WHAT_TO_INSPECT__	__HOW_TO_RECOVER__

12 Mental Model

Why this matters: __ONE_SHORT_VIVID_ANALOGY_THE_READER_WILL_REMEMBER_IN_SIX_MONTHS__

13 What You Should Understand Now

- __COMPREHENSION_POINT_ONE__
- __COMPREHENSION_POINT_TWO__

14 Next Step

__NEXT_GUIDE_OR_EXERCISE_BY_NAME__